# 1 Proof that the Galois group of any irreducible factor of $x^n + ax + b$ is symmetric for different small values of $a$ and $b$

## 1.1 Summary of results

Fix $n > 0$ and $a, b \neq 0$ and let

$$f = x^n + ax + b$$

for legacy reasons define also

$$f_{n,a} = x^n + ax + 1$$

If all the conditions in one of the rows of the table below are satisfied, then the Galois group of any irreducible factor of $f$ is the full symmetric group.

| $a =$ | $b =$ | $n \not\equiv$ | mod |
|---|---|---|---|
| $\pm 1$ | $\pm 1$ | | |
| $\pm 2$ | $1$ | $2$ | $4$ |
| $\pm 3$ | $1$ | $-a$ | $18$ |
| $\pm 4$ | $1$ | $4$ | $8$ |
| $5$ | $1$ | $10, 40, 45, 60, 90, 95$ | $100$ |
| $-5$ | $1$ | $5, 10, 40, 55, 60, 90$ | $100$ |

## 1.2 Notations

Fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ and for each prime $p$, fix an algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ and an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$ extending the natural embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$. Denote by $G$ the absolute Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$, and for each prime $p$, denote by $I(p) \subseteq G$ the absolute inertia group at $p$, i.e., the embedding into $G$ of the inertia group $\mathrm{Gal}\left(\overline{\mathbb{Q}_p}/\mathbb{Q}_p^{\mathrm{un}}\right)$, where $\mathbb{Q}_p^{\mathrm{un}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}_p}$. For each polynomial $g$ and number field $K$, we will denote by $\mathrm{d}(g)$ and $\mathrm{d}(K)$ (or for legacy reasons also $\mathrm{disc}(K)$) the discriminant of $g$ and of $K$ respectively and by $\mathrm{rd}(K)$ the root discriminant of $K$, i.e. $\mathrm{rd}(K) = \mathrm{d}(K)^{1/[K:\mathbb{Q}]}$.

## 1.3 Strategy

We will first lay down a few results that will provide us with a strategy to show that for any triple $n, a, b$ that satisfies different conditions, the Galois group of any factor of $f$ is symmetric.

**Lemma 1.1.** *For any $p \nmid ab$, the action of the inertia group $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(f), \overline{\mathbb{Q}}\right)$ has at most two non-fixed points.*

I think I saw somewhere a proof that covers this for the irreducible case, but I can't find it now. So I will try to adapt Martin's proof

*Proof.* If $I(p)$ acts trivially, we are done. Assume that it does not. It suffices to show that the roots of $f$ has a single orbit under $I(p)$, and that the length of this orbit is 2. If two roots are in the same orbit under $I(p)$, then they have the same reduction modulo $p$. Therefore, if we denote by $\overline{f} \in \mathbb{F}_p[x]$ the reduction of $f$ modulo $p$, then it suffices to show that $f$ has exactly one root in $\overline{\mathbb{F}_p}$ of multiplicity larger than 1 and that its multiplicity is in fact 2.

Firstly, note that by [10, Theorem 2]

$$\mathrm{d}(f) = (-1)^{\frac{n(n-1)}{2}} \left(n^n b^{n-1} + (1-n)^{n-1} a^n\right)$$

and hence, since $p|\mathrm{d}(f)$ and since $p \nmid ab$, it follows that $p$ divides $n$ iff it divides $(n-1)$ and thus it divides neither.

Next, since

$$f' = nx^{n-1} + a$$
$$f'' = n(n-1)x^{n-2}$$

and since $p \nmid n(n-1), a$ it follows that $f', f''$ have no shared root in $\overline{\mathbb{F}_p}$ and hence $f$ has no triple roots in $\overline{\mathbb{F}_p}$.

Finally, note that any double root of $f$ is also a solution of the linear equation

$$g := nf - xf' = nx^n + nax + nb - nx^n - ax = a(n-1)x + bn$$

and hence since $p$ does not divide $a(n-1)$ there is exactly one double root of $f$ in $\overline{\mathbb{F}_p}$. $\square$

**Corollary 1.2.** *Let $g$ be an irreducible factor of $f$, then for any $p \nmid ab$, the action of the inertia group $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right)$ has at most two non-fixed points.*

*Proof.* Let

$$\pi : \mathbb{Q}[x] / (f) \to \mathbb{Q}[x] / (g)$$

be the quotient map. Then the precomposition map

$$\pi^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x] / (g), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x] / (f), \overline{\mathbb{Q}}\right)$$

is a monomorphism of $G$-sets and hence the result follows from the previous lemma - 1.1. $\square$

**Corollary 1.3.** *Any number field $K$ that embeds non-surjectively into $\mathbb{Q}[x] / (g)$ for any irreducible factor $g$ of $f$ is ramified only over primes dividing $ab$.*

*Proof.* It suffices to show that $K$ is unramified above any prime $p$ that does not divide $ab$. Let $p$ be such a prime. It suffices to show that $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$. Choose an embedding $\iota : K \hookrightarrow \mathbb{Q}[x] / (g)$. Then the precompositio map

$$\iota^* : \mathrm{Hom}\left(\mathbb{Q}[x] / (g), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective $G$-equivariant map. Thus, $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$

since any non-trivial action of $I(p)$ on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$ would lift to an action on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right)$ that would have at least 4 non-fixed points, contradicting Corollary 1.2. $\square$

**Lemma 1.4.** *A transitive permutation group that is generated by transpositions is the full symmetric group.*

*Proof.* Sketch inspired by a discussion with Itamar Israeli:

There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)(a\ b)(b\ c) = (a\ c)$$

$\square$

**Corollary 1.5.** *A primitive permutation group that contains a transposition is the full symmetric group.*

*Proof.* Slightly inspired by Borys's thesis:

Let $H$ be a primitive permutation group that contains a transposition $\tau$. Let $K$ be the normal closure of $\langle\tau\rangle$ inside $H$. Then the orbits of $K$ form a system of blocks for $H$. Therefore, $K$ is itself transitive and it is generated by transpositions and hence by the previouse lemma $K = S_n$ and hence $H = S_n$ as required. $\square$

**Corollary 1.6.** *If the galois group $G_L$ of a field $L$ of degree $n$ contains a transposition and $L$ has no proper intermediate extensions, then $G_L = S_n$.*

*Proof.* According to Corollary 1.5 it suffices to show that $G_L$ is primitive. Assume that it is not. Then there is a strictly smaller non-trivial $G$ set $S$ and a surjective $G$-equivariant map

$$\pi : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}\left(\alpha\right), \overline{\mathbb{Q}}\right) \to S$$

In particular, $\pi$ is an epimorphism of $G$-sets, and hence by the famous equivalnece of categories between $G$-sets and Etale algebras over $\mathbb{Q}$, $\pi$ corresponds to a monomorphism of Etale algebras over $\mathbb{Q}$, i.e. a field embedding

$$\iota : K \hookrightarrow L$$

and since $S$ is non-trivial and strictly smaller, $K$ is a proper subextension of $L/\mathbb{Q}$, contradicting the assumption that $L$ has no proper intermediate extensions. $\square$

**Corollary 1.7.** *If $g$ is an irreducible factor of $f$, there is a prime that divides the discriminant of $\mathbb{Q}[x]/(g)$ but does not divide $ab$, and $\mathbb{Q}[x]/(g)$ has no proper intermediate extensions, then the Galois group of $g$ is $S_{\deg g}$.*

*Proof.* Let $p$ be a prime that divides the discriminant of $\mathbb{Q}[x]/(g)$ but does not divide $ab$. Then by Corollary 1.2, $I(p)$ acts on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g), \overline{\mathbb{Q}}\right)$ as a single transposition and hence the Galois group of $g$ contains a transposition. The result the follows from Corollary 1.6. $\square$

**Corollary 1.8.** *Let $g$ be an irreducible factor of $f$, and assume that for every subfield $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x]/(g)$ there is a prime that does not divide $ab$ over which $K$ ramifies, then the Galois group of $g$ is $S_{\deg g}$.*

*Proof.* According to Corollary 1.3, $\mathbb{Q}[x]/(g)$ has no non-trivial subfields, and hence, since its discriminant is divisible by a prime that does not divide $a$, the Galois group of $g$ is $S_{\deg g}$ by Corollary 1.7. $\square$

Our strategy would be to find conditions on $n, a, b$ that would guarantee the conditions of the last corollary for any irreducible factor $g$ of $f$.

## 1.4 $ab = \pm 1$

Assume that $ab = \pm 1$.

**Corollary 1.9.** *The Galois group over $\mathbb{Q}$ of any irreducible factor of $f$ is symmetric.*

**Remark 1.10.** *According to Selmer [8] $f$ can always be reduced to either of the forms*

$$x^n - x - 1 \quad \textit{or} \quad x^n + x + 1$$

*by substituting $x$ by $-x$. Moreover he showed that the first form is always irreducible, while the second form is irreducible whenever $n \not\equiv 2 \mod 3$ and factors into $x^2+x+1$ times an irreducible factor otherwise. When $x^n \pm x + 1$ is irreducible, the symmetry of its Galois group follows for example from [7, Theorem 1]. I did not find a reference for the reducible case.*

*Proof.* This is immediate from Corollary 1.8 since there is no prime that divides $ab \pm 1$ and the discriminant of any non-trivial extension of $\mathbb{Q}$ is divisible by some prime as a result of the Minkowski bound. $\square$

## 1.5 $a = \pm 2, b = 1$

We will handle the case $a = -2$ first. Note that in this case 1 is a root of $f_{n,-2}$, and if we divide $f_{n,-2}$ by $x - 1$ we obtain the polynomial

$$g := x^{n-1} + x^{n-2} + \ldots + x - 1 \tag{1}$$

After substituting $x$ by $1/x$ and multiplying by $-x^{n-1}$ we obtain the polynomial

$$x^{n-1} - x^{n-2} - \ldots - x - 1$$

which defines the same ring extension over $\mathbb{Q}$ as $g$ and which is none other than the characteristic polynomial of the generalized Fibonacci sequence which was studied by Wolfram [11]. Wolfram showed that this polynomial is irreducible over $\mathbb{Q}$, and Martin [3] showed that whenever $n$ is odd, its Galois group is the full symmetric group. We will show that its Galois group is the full symmetric group for any $n$. However,

we will stick to our version of the polynomial defined in (1).

We will denote by $d$ the discriminant of $g$. Then:

**Lemma 1.11.**

$$v_2(d) = \begin{cases} n-2 & n \text{ is even} \\ 0 & n \text{ is odd} \end{cases}$$

*Proof.* By [3]

$$d = \frac{(-1)^{\binom{n}{2}} \left(n^n - 2^n (n-1)^{n-1}\right)}{(n-2)^2}$$

If $n$ is odd it is easy to see that both the denominator and numerator are odd, and hence $v_2(d) = 0$ as required. Assume now that $n$ is even. Then we can manipulate the expression above and get

$$d = (-1)^{\binom{n}{2}} 2^n \cdot \frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{(n-2)^2}$$

and hence if we denote $m = v_2(n-2)$ we get that

$$v_2(d) = n - 2m + v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right)$$

Therefore, our claim is equivalent to showing that

$$n-2=n-2m+v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)$$

or equivalently, that

$$v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)=2m-2$$

Assume first that $m=1$, then

$$v_2\left(\frac{n}{2}\right)\geq 1$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n\right)\geq n>0=v_2\left((n-1)^{n-1}\right)$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n-(n-1)^{n-1}\right)=0=2m-2$$

Assume now that $m\geq 2$. In the rest of the proof we will use greek letters solely to denote different odd integer whose exact value is unknown and immaterial. As a first example for this notation, we

denote

$$n-2 \equiv \epsilon 2^m \mod 2^{2m-1}$$

From this we may deduce that

$$\begin{aligned}
&\left(\frac{n}{2}\right)^n - (n-1)^{n-1} \equiv \\
&\left(\frac{\epsilon 2^m + 2}{2}\right)^n - (\epsilon 2^m + 1)^{n-1} \equiv \\
&\left(\eta 2^{m-1} + 1\right)^n - (\epsilon 2^m + 1)^{n-1} \equiv \backslash 3m - 3 \geq 2m - 1 \\
&\binom{n}{2} \eta^2 2^{2m-2} + (n+1)\,\epsilon 2^{m-1} + 1 - (n-1)\,\epsilon 2^m - 1 \equiv \\
&\frac{(\epsilon 2^m + 2)\,(\epsilon 2^m + 1)}{2} \eta^2 2^{2m-2} + (\epsilon 2^m + 2)\,\epsilon 2^{m-1} - (\epsilon 2^m + 1)\,\epsilon 2^m \equiv \\
&\iota \eta^2 2^{2m-2} + \epsilon^2 2^{2m-1} + \epsilon 2^m - \epsilon^2 2^{2m} - \epsilon 2^m \equiv \\
&2^{2m-2}\left(\iota \eta^2 - 2\epsilon^2\right) \\
&2^{2m-2}\alpha \mod 2^{2m-1}
\end{aligned}$$

and in particular its 2-adic valuation is $2m-2$ as required. □

**Lemma 1.12.** *Let $p$ be a prime and $L$ a number field. Then*

$$v_p\left(\mathrm{rd}K\right) \leq v_p\left(\mathrm{rd}L\right)$$

*Proof.* Indeed,

$$\operatorname{disc}(L) = \operatorname{disc}(K)^{[L:K]} \operatorname{Norm}_{K/\mathbb{Q}}(\operatorname{disc}(L/K))$$

and hence

$$\begin{aligned}
&v_p(\operatorname{disc}(L)) = \\
&[L:K]\, v_p(\operatorname{disc}(K)) + v_p\left(\operatorname{Norm}_{K/\mathbb{Q}}(\operatorname{disc}(L/K))\right) \geq \\
&[L:K]\, v_p(\operatorname{disc}(K))
\end{aligned}$$

and hence

$$v_p(\mathrm{d}(K)) \leq v_p(\mathrm{d}(L)) / [L:K]$$

$$\begin{aligned}
&v_p(\mathrm{rd}K) = \\
&\frac{v_p(\operatorname{disc}(K))}{[K:\mathbb{Q}]} \leq \\
&\frac{v_p(\operatorname{disc}(L)) / [L/K]}{[K:\mathbb{Q}]} = \\
&\frac{v_p(\operatorname{disc}(L))}{[L:\mathbb{Q}]} = \\
&v_p(\mathrm{rd}L)
\end{aligned}$$

□

**Corollary 1.13.** *The Galois group of the non-linear factor of* $x^n - 2x + 1$ *is* $S_{n-1}$.

*Proof.* Let $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x]/(g)$ be a sub-extension. Then according to Lemma 1.12

$$v_2(\mathrm{rd}K) \leq v_2(\mathrm{rd}(\mathbb{Q}[x]/(g))) \leq \frac{v_2(g)}{n-1} \leq \frac{n-2}{n-1} < 1$$

Therefore, if $\mathrm{d}(K)$ is divisible only by 2, then $\mathrm{rd}(K) < 2$. However, theere are no number fields with root discriminant smaller than 2 that are ramified only over 2 we can prove it either with the Minkowski bound as I did in the first version of the proof, or just by citing `https://www.lmfdb.org/NumberField/?rd=-2-2` like we do in the rest of the cases . Therefore, the discriminant of $K$ is divisible by a prime that does not divide $a$ and hence the Galois group of $g$ is $S_{n-1}$ by Corollary 1.8. □

**Lemma 1.14.** *Assume that* $n$ *is odd. Then* $x^n + 2x + 1$ *is irreducible over* $\mathbb{Q}$.

*Proof.* First of all note that $x^n + 2x + 1$ has no roots on the unit circle in $\mathbb{C}$. Indeed, if $x$ is such a root, then

$$1 = |2x + 1| \geq |2x| - |1| = 1$$

and hence $x = -1$. However,

$$(-1)^n + 2(-1) + 1 = -1 - 2 + 1 = -2 \neq 0$$

Consider now the polynomial

$$x^n \left((1/x)^n + 2(1/x) + 1\right) = x^n + 2x^{n-1} + 1$$

whose roots are exactly the inverses of the roots of $x^n + 2x + 1$. For any small enough $\epsilon > 0$, the winding number of $x^n + 2x^{n-1} + 1$ around 0 as $x$ traverses the circle of radius $1 + \epsilon$ is $n - 1$. Indeed, for such $\epsilon$,

$$|2x^{n-1}| > |x^n| + 1 \geq |x^n + 1|$$

since

$$(1 + \epsilon)^n - 2(1 + \epsilon)^{n-1} + 1 < 0$$

since

$$\frac{d}{dx}\left(x^n - 2x^{n-1} + 1\right)|_{x=1} = n - 2\left(n-1\right) = 2 - n < 0$$

However, since $x^n + 2x + 1$ has no roots on the unit circle, and hence so does $x^n + 2x^{n-1} + 1$, it follows that all of these $n-1$ roots are in the interior of the unit circle. In other words, only one of its roots has absolute value greater than or equal to 1 (in fact greater). If we assume towards contradiction that $x^n + 2x^{n-1} + 1$ is reducible over $\mathbb{Q}$, then since it is monic, it is reducible over $\mathbb{Z}$ as well, and hence one of its factors is a monic polynomial with integer coefficients whose roots are all of absolute value strictly less than 1 which is a contradiction (because then for example the norm, which is the free coefficient up to a sign, is less than 1 in absolute value). Therefore, $x^n + 2x^{n-1} + 1$ is irreducible over $\mathbb{Q}$ and hence so is $x^n + 2x + 1$ as required. $\square$

**Corollary 1.15.** *$x^n + 2x + 1$ is irreducible whenever $n$ is odd and factors into $x + 1$ times an irreducible factor otherwise. The Galois group of any irreducible factor $g$ of $f = x^n + 2x + 1$ is $S_{\deg g}$.*

*Proof.* Assume first that $n$ is even, then

$$(-x)^n + 2(-x) + 1 = x^n - 2x + 1$$

and hence by the above analysis, $x^n - 2x + 1$ factors into $x + 1$ times an irreducible factor whose Galois group is $S_{n-1}$.

Assume now that $n$ is odd. According to [10, Theorem 2]

$$\mathrm{d}(f_{n,2}) = (-1)^{\frac{n(n-1)}{2}} \left(n^n + (1-n)^{n-1} 2^n\right)$$

and hence $v_2(\mathrm{d}(f_{n,2})) = 0$. Therefore, discriminant of any intermediate extension $\mathbb{Q} \subsetneq K \subseteq \mathbb{Q}[x]/(f_{n,2})$ is, odd and in particular it is divisible by a prime that does not divide 2 and hence the Galois group of $f_{n,2}$ is $S_n$ by Corollary 1.8. $\square$

## 1.6 $a = \pm 3, \pm 4, \pm 5$

**Lemma 1.16.** *If $|a| > 2$ then $f_{n,a}$ is irreducible over $\mathbb{Q}$.*

*Proof.* It suffices to show that

$$\widehat{f_{n,a}} := x^n f_{n,a}(1/x) = x^n + ax^{n-1} + 1$$

is irreducible over $\mathbb{Q}$. For any $x \in S^1 \subseteq \mathbb{C}$,

$$\left|ax^{n-1}\right| = |a| > 2 = |x^n| + |1| \geq |x^n + 1|$$

and hence the winding number of $\widehat{f_{n,a}}(x)$ around 0 as $x$ traverses the unit circle is equal to that of $-ax^{n-1}$ which is $n-1$. Therefore, $\widehat{f_{n,a}}$ has $n-1$ roots in the interior of the unit disk and hence it is irreducible over $\mathbb{Q}$. $\square$

**Remark 1.17.** *The above lemma also follows from [2, Theorem 1].*

**Lemma 1.18.** *Let $p$ be a prime dividing $a$. If*

$$v_p(n) \neq v_p(a) \;\; or \;\; (n/a)^n \not\equiv -1 \mod p$$

*then*

$$v_p(disc(f_{n,a})) = n \min\{v_p(n), v_p(a)\}$$

*Proof.* Assume first that $v_p(n) \neq v_p(a)$. According to [10, Theorem 2]

$$\operatorname{disc}(f_{n,a}) = (-1)^{\frac{n(n-1)}{2}} \left(n^n + (1-n)^{n-1} a^n\right)$$

Then either $v_p(n) = 0$ and hence $v_p(\operatorname{disc}(f_{n,a})) = 0$ or $v_p(n) > 0$ and

hence $v_p(n-1)=0$ and thus $v_p\left((1-n)^{n-1}a^n\right)=nv_p(a)$ and hence

$$v_p(\operatorname{disc}(f_{n,a}))=\min\left\{v_p(n^n),v_p\left((1-n)^{n-1}a^n\right)\right\}=n\min\{v_p(n),v_p(a)\}$$

Assume now that $v_p(n)=v_p(a)$. Then

$$\operatorname{disc}(f_{n,a})=\operatorname{Res}\left(f_{n,a},f'_{n,a}\right)=p^{nv_p(a)}\operatorname{Res}\left(f_{n,a},p^{-v_p(a)}f'_{n,a}\right)$$

Since $p^{-v_p(a)}f'_{n,a}\in\mathbb{Z}[x]$ it follows that $v_p(\operatorname{disc}(f_{n,a}))\geq nv_p(a)$ and equality holds iff $f_{n,a}$ and $p^{-v_p(a)}f'_{n,a}$ have no common root in $\overline{\mathbb{F}_p}$. Assume then that there is no equality, i.e. that

$$\begin{aligned}\overline{f_{n,a}}&=x^n+ax+1=x^n+1\in\mathbb{F}_p[x]\\ p^{-v_p(a)}\overline{f_{n,a}}'&=np^{-v_p(a)}x^{n-1}+ap^{-v_p(a)}\in\mathbb{F}_p[x]\end{aligned}$$

have a common root. This root in particulat annahilates

$$g_{n,a}:=x\overline{f_{n,a}}'-n\overline{f_{n,a}}=nx^n+ax-nx^n-n=ax-n$$

i.e. it is equal to $n/a$ and hence

$$0 = \overline{f_{n,a}}\left(n/a\right) = \left(n/a\right)^n + 1$$

and thus $\left(n/a\right)^n \equiv -1 \mod p$. $\square$

Assume in the sequel that $|a| < 22$ (or $< 44$ assuming GRH). According to [6], there is a constant $C\left(|a|\right)$ such that for any number field $K$ with degree $m > C\left(|a|\right)$,

$$\left|\operatorname{disc}\left(K\right)^{1/m}\right| > |a|$$

Moreover, let $S\left(|a|\right)$ be the set of positive integers $n$ that satisfy the conditions of Lemma 1.18, i.e. for any prime $p \mid a$ either

$$v_p\left(n\right) \neq v_p\left(a\right) \text{ or } \left(n/a\right)^n \not\equiv -1 \mod p$$

**Lemma 1.19.** *Let $C\left(|a|\right) < n \in S\left(|a|\right)$. Then there is a prime $p$ s.t. $p \mid disc\left(\mathbb{Q}\left(\alpha\right)\right)$ but $p \not| a$ and consequently $\mathbb{Q}\left(\alpha\right)$ ramifies above $p$.*

The only example I found so far where every prime dividing the discriminant also divides $a$ is $n = 3, a = -3$ and I checked for any

$n \leq 25$ and $|a| \leq 22$.

*Proof.* According to Lemma 1.18 for any prime $p$ dividing $a$

$$v_p\left(\operatorname{disc}\left(\mathbb{Q}\left(\alpha\right)\right)\right) \leq v_p\left(\operatorname{disc}\left(f_{n,a}\right)\right) = n \min\left\{v_p\left(n\right), v_p\left(a\right)\right\} \leq n v_p\left(a\right)$$

and hence if any prime dividing $n$ also divides $a$ then

$$\left|\operatorname{disc}\left(\mathbb{Q}\left(\alpha\right)\right)\right| \leq \prod_{p|a} p^{n v_p(a)} = |a^n| \leq 22^n$$

However, according to [5] for any large enough $n$

$$\left|\operatorname{disc}\left(\mathbb{Q}\left(\alpha\right)\right)\right| > 22^n$$

(or $> 44^n$ assuming GRH) and hence there is a prime dividing $\operatorname{disc}\left(\mathbb{Q}\left(\alpha\right)\right)$ that does not divide $a$. The consequence follows from Lemma 1.1. $\square$

**Lemma 1.20.** *Assume that $|a| > 2$ and that $n \in S\left(|a|\right)$. Then for any intermediate extension $\mathbb{Q}\left(\alpha\right)/K/\mathbb{Q}$ of degree $m$,*

$$\left|disc\left(K\right)^{1/m}\right| \leq |a|$$

*Proof.* Indeed,

$$\operatorname{disc}(\mathbb{Q}(\alpha)) = \operatorname{disc}(K)^{[\mathbb{Q}(\alpha):K]} \operatorname{Norm}_{K/\mathbb{Q}}(\operatorname{disc}(\mathbb{Q}(\alpha)/K))$$

and hence for any prime $p$ dividing $\operatorname{disc}(K)$,

$$\frac{n}{m} v_p(\operatorname{disc}(K)) \leq v_p(\operatorname{disc}(\mathbb{Q}(\alpha))) \leq v_p(\operatorname{disc}(f_{n,a})) = n \min\{v_p(n), v_p(a)\} \leq n v_p(a)$$

and hence

$$\frac{v_p(\operatorname{disc}(K))}{m} \leq v_p(a)$$

Since it is true for any prime dividing $\operatorname{disc}(K)$, it follows that

$$\left|\operatorname{disc}(K)^{1/m}\right| \leq |a|$$

□

**Lemma 1.21.** *Let $n \in S(|a|)$. Assume that there is no intermediate extension $\mathbb{Q} \subsetneq K \subsetneq \mathbb{Q}(\alpha)$ that is ramified only over primes dividing $a$ and whose root discriminant is at most $|a|$. Then $Gal(f_{n,a}) = S_n$.*

*Proof.* Assume that $n$ satisfies the above conditions. Then by Lemma

1.20, $\mathbb{Q}(\alpha)$ has no proper intermediate extensions. Therefore, $\mathrm{Hom}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ has no $G$-equivariant quotients and hence $G$ acts primitively on $\mathrm{Hom}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$. However, by Lemma 1.1 and Lemma 1.19, there is a prime $p$ such that $I(p)$ acts on $\mathrm{Hom}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ as a single transposition and hence $G$ contains a transposition. Therefore, since any primitive subgroup of $S_n$ that contains a transposition is $S_n$, it follows that $G = S_n$. $\square$

**Lemma 1.22.** *Let $K$ be a number field. Assume that there is an integer $m \in \mathbb{Z}$ and a prime $p$ not dividing $a$ such that $K$ has no embedding into $\mathbb{Q}_p$ and*

$$v_p\left(f_{n,a}(m)\right) = 1$$

*Then $K$ cannot be a subfield of $\mathbb{Q}[x] / (f_{n,a})$.*

*Proof.* Since $p$ does not divide $a$, it does not ramify in $K$ according to Corollary 1.3, and hence since $K$ has no embedding into $\mathbb{Q}_p$, $K$ has no embedding into any purely ramified extension of $\mathbb{Q}_p$. Therefore, to show that $K$ can not be a subfield of $\mathbb{Q}[x] / (f_{n,a})$, it suffices to show that $f_{n,a}$ has a root that is contained in a purely ramified extension of $\mathbb{Q}_p$. Indeed, let $e$ be the multiplicity of $m$ as a root of $f_{n,a}$ modulo $p$.

Then there is a factorization

$$\pi\left(f_{n,a}\right)=(x-m)^{e}\,\overline{g}\in\mathbb{F}_{p}\left[x\right]$$

for some $\overline{g}\in\mathbb{F}_{p}\left[x\right]$ that is coprime to $(x-m)$ where $\pi:\mathbb{Z}_{p}\left[x\right]\rightarrow\mathbb{F}_{p}\left[x\right]$ is the quotient map. By hensel's lemma, there is a factorization

$$f_{n,a}=hg\in\mathbb{Z}_{p}\left[x\right]$$

where $\pi\left(h\right)=(x-m)^{e}$ and $\pi\left(g\right)=\overline{g}$. Consider now the polynomial

$$f_{n,a}\left(x+m\right)=h\left(x+m\right)g\left(x+m\right)=x^{n}+\sum_{i=0}^{n-1}c_{i}x^{i}\in\mathbb{Z}\left[x\right]$$

Then

$$v_{p}\left(c_{0}\right)=v_{p}\left(f_{n,a}\left(m\right)\right)=1$$

and

$$\pi\left(f_{n,a}\left(x+m\right)\right)=\pi\left(h\left(x+m\right)\right)\pi\left(g\left(x+m\right)\right)=x^{e}\overline{g}\left(x+m\right)\in\mathbb{F}_{p}\left[x\right]$$

and hence

$$v_p(c_i) \geq 1$$

for all $i < e$. Therefore, the Newton polygon of $f_{n,a}(x+m)$ with respect to $p$ has a single segment of negative and its slope is at most $1/e$ in absolute value. Therefore, $f_{n,a}(x+m)$ has a root, call it $\beta$, whose $p$-valuation is at most $1/e$. Therefore, since the set of roots of $f_{n,a}(x+m)$ that reduce to 0 modulo $p$ is exactly the set of roots of $h(x+m)$, whose degree is $e$, it follows that the ramification degree of $\beta$ over $\mathbb{Q}_p$ is exactly its over all algebraic degree over $\mathbb{Q}_p$ and hence $\mathbb{Q}_p(\beta)/\mathbb{Q}_p$ is purely ramified. This shows that there is an embedding, in particular the one extending the assignment $x \mapsto \beta$, of $\mathbb{Q}[x]/(f_{n,a})$ into a purely ramified extension of $\mathbb{Q}_p$ $\square$

**Corollary 1.23.** *Assume that that for every number field $K$ that is ramified only over primes dividing $a$ and whose root discriminant is at most $|a|$, either of the following holds:*

1. *$K$ is purely imaginary.*

2. *There is a prime $p$ not divide $a$ such that $v_p(a+2) = 1$ and such that there is a polynomial $h$ defining $K$ over $\mathbb{Q}$ that has no solution*

*mod p.*

3. *$K$ is of even degree and there is a prime $p$ such that $v_p(-a+2)=1$ and such that there is a polynomial defining $K$ over $\mathbb{Q}$ that has no solution mod $p$.*

*Then $Gal(f_{n,a}) = S_n$ for any $C(|a|) < n \in S(|a|)$.*

*Proof.* Based on Lemma 1.21, it suffices to show that any field $K$ that satisfies one of the above conditions cannot be a subfield of $\mathbb{Q}[x]/(f_{n,a})$.

1. If $k$ is purely imaginary, note that $f_{n,a}$ always has a real root as it tends to $+\infty$ as $x \to +\infty$ and $f_{n,a}(a/|a|)$ is always negative.

2. Assume that the second condition holds. Since $h$ has no solution mod $p$, $K$ has no embedding into $\mathbb{Q}_p$ and hence since

$$1 = v_p(a+2) = v_p(f_{n,a}(1))$$

   it follows from Lemma 1.22 that $K$ cannot be a subfield of $\mathbb{Q}[x]/(f_{n,a})$.

3. Assume that the third condition holds. If $n$ is odd, then $K$ cannot be a subfield of $\mathbb{Q}[x]/(f_{n,a})$ since the latter has an odd degree.

Assume then that $n$ is even. Again, since $h$ has no solution mod $p$, $K$ has no embedding into $\mathbb{Q}_p$ and hence since

$$1 = v_p\left(-a+2\right) = v_p\left(f_{n,a}\left(-1\right)\right)$$

it follows from Lemma 1.22 that $K$ cannot be a subfield of $\mathbb{Q}\left[x\right]/\left(f_{n,a}\right)$.

□

We can now use calculated minimal root discriminants of number fields ramified only over primes dividing $a$ to deduce which values of $a$ satisfy the condition of lemma 1.21.

**Corollary 1.24.** *For any $n \in S\left(3\right)$, $Gal\left(f_{n,\pm 3}\right) = S_n$ unless $n = 3, a = -3$.*

*Proof.* According to `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=3&rd=-3-3`, the only non-trvial number field that is ramified only over 3 and whose root discriminant is at most 3 is the cyclotomic extension $\mathbb{Q}\left(\zeta_3\right) = \mathbb{Q}\left(\sqrt{-3}\right)$ which is purely imaginary and hence the result follows from Corollary 1.23 for any $n > C\left(3\right)$. However, according to `https://www.lmfdb.org/NumberField/?ram_`

quantifier=subset&rd=-5-5&sort_order=rd&showcol=rd&hidecol=galois.class_group_desc&sort_order=rd, $C(3) = 3$ and the smaller cases can be checked computationally. □

**Corollary 1.25.** *For any* $n \in S(4)$, $Gal(f_{n,\pm 4}) = S_n$.

*Proof.* According to https://www.lmfdb.org/NumberField/?ram_quantifier=subset&ram_primes=2&rd=-4-4&sort_order=rd&sort_order=rd, the only non-trivial number fields that are ramified only over 2 and whose root discriminant is at most 4 are the three purely imaginary fields defined by the polynomials $x^2+1, x^2+2, x^4+1$, and the quadratic field $\mathbb{Q}\left(\sqrt{2}\right)$ which is defined by $x^2-2$ which has no solution mod 3 which divides $4+2 = -(-4)+2$, is inert, and hence the result follows from Corollary 1.23 for any $n > C(4)$. However, according to https://www.lmfdb.org/NumberField/?ram_quantifier=subset&rd=-5-5&sort_order=rd&showcol=rd&hidecol=galois.class_group_desc&sort_order=rd, $C(4) = 4$ and the smaller cases can be checked computationally. □

**Corollary 1.26.** *For any* $n \in S(5)$, $Gal(f_{n,\pm 5}) = S_n$.

*Proof.* According to https://www.lmfdb.org/NumberField/?ram_quantifier=

`subset&ram_primes=5&rd=-5-5&sort_order=rd&sort_order=rd`, the only non-trivial number fields that are ramified only over 5 and whose root discriminant is at most 5 are the cyclotomic extension $\mathbb{Q}(\zeta_5)$ and the quadratic field $\mathbb{Q}\left(\sqrt{5}\right)$. The first is purely imaginary, and the second may be defined by $x^2-5$ which has no solution either mod 7 which divides $5+2$ or mod 3 which divides $-5+2$ and hence the result follows from Corollary 1.23 for any $n > C(5)$. However, according to `https://www.lmfdb.org/NumberField/?ram_quantifier=subset&rd=-5-5&sort_order=rd&showcol=rd&hidecol=galois.class_group_desc&sort_order=rd`, $C(5) = 6$ and the smaller cases can be checked computationally. □

I cannot reliably check any largegr values since the database is guaranteed to be complete only for root discriminants up to 5.989.

The following table summerizes the set $S(a)$ for each $a = \pm 3, \pm 4, \pm 5$:

| $a$ | $n \in S(a) \Leftrightarrow n \not\equiv$ | mod |
|---|---|---|
| 2 | 2 | 4 |
| $\pm 3$ | $-a$ | 18 |
| $\pm 4$ | 4 | 8 |
| 5 | $10, 40, 45, 60, 90, 95$ | 100 |
| $-5$ | $5, 10, 40, 55, 60, 90$ | 100 |

**Remark 1.27.** *The cases where $n$ and $a$ are coprime, follow from [7, Theorem 1]. For the other cases I did not find any published proof.*

# 2 Complete Proof

The indices of $f_n$ has a shift by one from the indices in the rest of the files in the repository

For each $n$ denote

$$f_n = x^n - x^{n-1} - \ldots - 1$$

It has been shown that $f_n$ is irreducible over $\mathbb{Q}$ ( I will add a citation later ). We will show that for each $n$, the Galois group of $f_n$ is

isomorphic to $S_n$.

Fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$, a root $\alpha \in \overline{Q}$ of $f_n$ and for each prime $p$, fix an algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ and an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$ extending the natural embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$. Denote by $G$ the absolute Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$, by $G_n$ the Galois group of $f_n$, and for each prime $p$, denote by $I(p) \subseteq G$ the absolute inertia group at $p$, i.e., the embedding into $G$ of the inertia group $\mathrm{Gal}\left(\overline{\mathbb{Q}_p}/\mathbb{Q}_p^{\mathrm{un}}\right)$, where $\mathbb{Q}_p^{\mathrm{un}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}_p}$.

Using this notation, we can first adapt part of Martin's proof of [3, Theorem 2.5] to show the following lemma:

**Lemma 2.1.** *For each odd prime $p$, $I(p)$ acts on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ either trivially or as a single transposition.*

*Proof.* postponed. □

In light of this lemma it makes sense to guess that the $G_n$ contains a transposition, since otherwise the discriminant of $\mathbb{Q}(\alpha)$ would be ramified only at 2. We can show this by analyzing the discriminant $d_n$ of $f_n$ for each $n$. However, before doing so, we will use the previous lemma to derive a lemma which, as we will explain afterwards, will

make the analysis of the discriminant even more useful:

**Lemma 2.2.** *Any proper subextension of* $\mathbb{Q}(\alpha)/\mathbb{Q}$ *is not ramified above any odd prime.*

*Proof.* Let $\iota : K \hookrightarrow \mathbb{Q}(\alpha)$ be a proper subextension and let $p$ be an odd prime. It suffices to show that any embedding of $K$ into $\overline{\mathbb{Q}_p}$ lands inside $\mathbb{Q}_p^{\mathrm{un}}$ for which it suffices to show that $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$. If $I(p)$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$ then this is immediate. Assume otherwise. Then by the previous lemma, $I(p)$ acts as a single transposition on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$. Moerover, the precomposition map

$$\iota^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective $G$-equivariant map. In particular it is

- Surjective with fibers of constant size $[\mathbb{Q}(\alpha) : K] \geq 2$;
- Compatible with the action of $I(p)$.

However, the only way a transposition can be compatible with such a map is if it swaps two elements inside a single fiber and thus acts

trivially on the image. $\square$

The reason this lemma makes the analysis of the discriminant even more useful, is that using the analysis of the discriminant, and using the Minkowski bound, we can in fact show that $\mathbb{Q}(\alpha)$ has no proper subextensions at all, from which we will derive that $G_n$ acts primitively on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$. This is in turn useful since a primitive permutation group that contains a transposition is, as we shall also show, the full symmetric group, and thus we will be able to conclude that $G_n \cong S_n$. Without further ado then, we bravely proceed to analyze the discriminant:

**Lemma 2.3.**

$$v_2(d_n) = \begin{cases} n-1 & n \text{ is odd} \\ 0 & n \text{ is even} \end{cases}$$

*Proof.* By [3]

$$d_n = \frac{(-1)^{\binom{n+1}{2}}\left((n+1)^{(n+1)} - 2^{n+1}n^n\right)}{(n-1)^2}$$

If $n$ is even it is easy to see that both the denominator and numerator are odd, and hence $v_2(d_n) = 0$ as required. Assume now that $n$ is odd.

Then we can manipulate the expression above and get

$$d_n = (-1)^{\binom{n+1}{2}} 2^{n+1} \cdot \frac{\left(\frac{n+1}{2}\right)^n - n^n}{(n-1)^2}$$

and hence if we denote $m = v_2(n-1)$ we get that

$$v_2(d_n) = n + 1 - 2m + v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right)$$

Therefore, our claim is equivalent to showing that

$$n - 1 = n + 1 - 2m + v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right)$$

or equivalently, that

$$v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right) = 2m - 2$$

Assume first that $m = 1$, then

$$v_2\left(\frac{n+1}{2}\right) \geq 1$$

and hence

$$v_2\left(\left(\frac{n+1}{2}\right)^n\right) \geq n > 0 = v_2\left(n^n\right)$$

and hence

$$v_2\left(\left(\frac{n+1}{2}\right)^n - n^n\right) = 0 = 2m-2$$

Assume now that $m \geq 2$. In the rest of the proof we will use greek letters solely to denote different odd integer whose exact value is unknown and immaterial. As a first example for this notation, we denote

$$n-1 \equiv \epsilon 2^m \mod 2^{2m-1}$$

From this we may deduce that

$$
\begin{aligned}
&\left(\frac{n+1}{2}\right)^{n+1} - n^n \equiv \\
&\left(\frac{\epsilon 2^m + 2}{2}\right)^{n+1} - \left(\epsilon 2^m + 1\right)^n \equiv \\
&\left(\eta 2^{m-1} + 1\right)^{n+1} - \left(\epsilon 2^m + 1\right)^n \equiv \backslash 3m - 3 \geq 2m - 1 \\
&\binom{n+1}{2} \eta^2 2^{2m-2} + (n+1)\,\epsilon 2^{m-1} + 1 - n\epsilon 2^m - 1 \equiv \\
&\frac{(\epsilon 2^m + 2)\,(\epsilon 2^m + 1)}{2} \eta^2 2^{2m-2} + (\epsilon 2^m + 2)\,\epsilon 2^{m-1} - (\epsilon 2^m + 1)\,\epsilon 2^m \equiv \\
&\iota \eta^2 2^{2m-2} + \epsilon^2 2^{2m-1} + \epsilon 2^m - \epsilon^2 2^{2m} - \epsilon 2^m \equiv \\
&2^{2m-2}\left(\iota \eta^2 - 2\epsilon^2\right) \\
&2^{2m-2}\alpha \mod 2^{2m-1}
\end{aligned}
$$

and in particular its 2-adic valuation is $2m - 2$ as required. □

A first easy corollary of this lemma is the following promised statement:

**Corollary 2.4.** *$G_n$ contains a transposition for each $n \geq 2$.*

*Proof.* This proof works only for even $n$ or $n$ that are sufficiently large. We could also cite [3, Theorem 2.5]. However, in his proof, he dismisses

pretty much all the content of the following proof as "easy to see", and I am not convinced yet.

By lemma 3.1, it suffices to show that there is an odd prime that ramifies in $\mathbb{Q}(\alpha)$, or equivalently, that divides the discriminant of $\mathbb{Q}(\alpha)$.

Assume first that $n$ is even this. Then by the previous lemma, $d_n$ is odd. However, since $1, \alpha, \ldots, \alpha^{n-1}$ spans a sublattice of the ring of integers of $\mathbb{Q}(\alpha)$, the discriminant of $\mathbb{Q}(\alpha)$ divides $d_n$ and is thus odd as well. Therefore, there is an odd prime , $p$, that divides the discriminant of $\mathbb{Q}(\alpha)$.

Assume now that $n$ is odd. By the same argument as before,

$$\operatorname{disc}(\mathbb{Q}(\alpha)) \,|\, d_n$$

and hence

$$v_2(\operatorname{disc}(\mathbb{Q}(\alpha))) \leq v_2(d_n) = n - 1$$

Therefore, it suffices to show that

$$\operatorname{disc}(\mathbb{Q}(\alpha)) > 2^{n-1}$$

By [5] for any sufficiently large $n$

$$\operatorname{disc}(\mathbb{Q}(\alpha)) > 22^n$$

so this is clearly true. I have not went through [5] carefully enough yet to see if they detail how large $n$ needs to be, so I am not sure how feasible it will be to check the remaining small values of $n$ computationally.

□

As promised, we can also use the previous lemma, to show that $\mathbb{Q}(\alpha)$ has no proper subextensions at all. While for even values of $n$ this is immediate, in order to derive it for odd values of $n$ we will first need to prove a lemma and two corolarries:

**Lemma 2.5.** *Let $K/\mathbb{Q}$ be a field extension of degree at least 3. Then*

$$disc(K) \geq 2^{[K:\mathbb{Q}]}$$

*Proof.* Denote $r = [K : \mathbb{Q}]$ and $d_K = \operatorname{disc}(K)$. It suffices to show that

$$\sqrt{d_K} \geq 2^{r/2}$$

However, by the Minkowski bound

$$\sqrt{d_K} \geq \left(\frac{\pi}{4}\right)^{r/2} \cdot \frac{r^r}{r!}$$

and hence it suffices to show that

$$\left(\frac{\pi}{4}\right)^{r/2} \cdot \frac{r^r}{r!} \geq 2^{r/2}$$

or equivalently that

$$\frac{r^r}{r!} \geq \left(\frac{8}{\pi}\right)^{r/2}$$

which is true for $r = 3$ and follows for larger $r$ by induction. $\square$

**Corollary 2.6.** *If* $[K : \mathbb{Q}] \geq 2$ *and* $K$ *is ramified only at* $2$ *then*

$$v_2\left(d_K\right) \geq [K : \mathbb{Q}]$$

*Proof.* If $r > 3$ it follows directly from the lemma. Assume that $r = 2$. Then $K = \mathbb{Q}\left(\sqrt{d}\right)$ for some integer $d$. Since $K$ is ramified only at 2, $d = \pm 2^k$ for some integer $k$. Therefore, one of three cases occures:

- $K = \mathbb{Q}\left(\sqrt{2}\right)$, in which case $d_K = 8$;

- $K = \mathbb{Q}\left(\sqrt{-2}\right)$, in which case $d_K = 8$;
- $K = \mathbb{Q}\left(\sqrt{-1}\right)$, in which case $d_K = 4$;

In each case $v_2\left(d_K\right) \geq 2$. $\square$

Even without analysing all the possible discriminants of quadratic extensions of $\mathbb{Q}$, we could use the previous lemma to show that $v_2\left(d_K\right) \geq [k : \mathbb{Q}] - 1$, and while the above corollary might seem only a slight improvement of this slightly weaker bound, the following corollary makes a very significant use of it:

**Corollary 2.7.** *There are no proper subextensions of* $\mathbb{Q}\left(\alpha\right)/\mathbb{Q}$.

*Proof.* Let $K \hookrightarrow \mathbb{Q}\left(\alpha\right)$ be a proper subextension. By Lemma 2.2, $K/\mathbb{Q}$ is ramified only at 2. Therefore, by the previous corollary,

$$v_2\left(d_K\right) \geq [K : \mathbb{Q}]$$

Then

$$d_{\mathbb{Q}(\alpha)} = d_K^{[\mathbb{Q}(\alpha):K]} \cdot N_{K/\mathbb{Q}}\left(d_{\mathbb{Q}(\alpha)/K}\right)$$

(see for example [9, III.§4]). Then

$$v_2\left(d_{\mathbb{Q}(\alpha)}\right) \geq [\mathbb{Q}(\alpha) : K] \cdot v_2(d_K)$$

However, $1, \alpha, \ldots, \alpha^{n-1}$ spans a sub lattice of the ring of integers of $\mathbb{Q}(\alpha)$ whose discriminant is $d_n$ and hence

$$n \leq v_2\left(d_{\mathbb{Q}(\alpha)}\right) \leq v_2(d_n) = n - 1$$

and that's a glaring contradiction. □

From this corollary, we can derive the following:

**Corollary 2.8.** *$G$ acts primitively on* $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$.

*Proof.* Assume towards contradiction that the action is imprimitive. Then there is a strictly smaller non-trivial $G$ set $S$ and a surjective $G$-equivariant map

$$\pi : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right) \to S$$

In particular, $\pi$ is an epimorphism of $G$-sets, and hence by the famous equivalnece of categories between $G$-sets and Etale algebras over $\mathbb{Q}$, $\pi$

corresponds to a monomorphism of Etale algebras over $\mathbb{Q}$, i.e. a field embedding

$$\iota : K \hookrightarrow \mathbb{Q}(\alpha)$$

and since $S$ is non-trivial and strictly smaller, $K$ is a proper subextension of $\mathbb{Q}(\alpha)/\mathbb{Q}$, contradicting the previous corollary. $\square$

It immediately follows that $G_n$ acts primitively on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}(\alpha), \overline{\mathbb{Q}}\right)$, and hence to completely seal the proof, it suffices to show the following lemma and a corollary:

**Lemma 2.9.** *A transitive permutation group that is generated by transpositions is the full symmetric group.*

*Proof.* Sketch inspired by a discussion with Itamar Israeli:

There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)(a\ b)(b\ c) = (a\ c)$$

□

**Corollary 2.10.** *A primitive permutation group that contains a transposition is the full symmetric group.*

*Proof.* Slightly inspired by Borys's thesis:

Let $H$ be a primitive permutation group that contains a transposition $\tau$. Let $K$ be the normal closure of $\langle\tau\rangle$ inside $H$. Then the orbits of $K$ form a system of blocks for $H$. Therefore, $K$ is itself transitive and it is generated by transpositions and hence by the previouse lemma $K = S_n$ and hence $H = S_n$ as required. □

# 3 Generalization to odd $k$

For each $n, k$ let

$$g_{n,k} = x^n - 2x^k + 1$$

Assume that $k$ is odd.

Fix an algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ and for each prime $p$, fix an algebraic closure $\overline{\mathbb{Q}_p}$ of $\mathbb{Q}_p$ and an embedding $\overline{\mathbb{Q}} \hookrightarrow \overline{\mathbb{Q}_p}$ extending the natural embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$. Denote by $G$ the absolute Galois group

$\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$, by $G_n$ the Galois group of $f_n$, and for each prime $p$, denote by $I(p) \subseteq G$ the absolute inertia group at $p$, i.e., the embedding into $G$ of the inertia group $\mathrm{Gal}\left(\overline{\mathbb{Q}_p}/\mathbb{Q}_p^{\mathrm{un}}\right)$, where $\mathbb{Q}_p^{\mathrm{un}}$ is the maximal unramified extension of $\mathbb{Q}_p$ inside $\overline{\mathbb{Q}_p}$.

Using this notation, we can first adapt part of Martin's proof of [3, Theorem 2.5] to show the following lemma:

**Lemma 3.1.** *For each odd prime $p$, $I(p)$ acts on $\mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(g_{n,k}), \overline{\mathbb{Q}}\right)$ either trivially or as a single transposition.*

*Proof.* If $I(p)$ acts trivially, we are done. Assume that it does not. It suffices to show that $I(p)$ acts on the roots of $g_{n,k}$ as a single transposition. If two roots are in the same orbit under $I(p)$, then they have the same reduction modulo $p$. Therefore, it suffices to show that $g_{n,k}$ has at most one pair of roots that are congruent modulo $p$, i.e. if we denote by $\overline{g_{n,k}} \in \mathbb{F}_p[x]$ the reduction of $g_{n,k}$ modulo $p$, then it suffices to show that $g_{n,k}$ has exactly one root in $\overline{\mathbb{F}_p}$ of multiplicity larger than 1 and that its multiplicity is in fact 2.

Firstly note that, any double root is also a root of

$$g'_{n,k} = nx^{n-1} - 2kx^{k-1}$$

and hence also a root of

$$h_{n,k} = ng_{n,k} - xg'_{n,k} = nx^n - 2nx^k + n - nx^n + 2kx^k = 2\left(k-n\right)x^k + n$$

$\square$

## 3.1 (a)

We will first show that $E\left(\mathbb{F}_q\right)$ has no point of order $p$. We have seen (proof of Corrolary $III.6.4.c$) that $\#E\left[p\right] = \deg_s\left(\hat{\phi}\right)$ when $\phi$ is the frobenius endomorphism of $E$ and $\hat{\phi}$ is its dual. Now $d$ the maximal order of a point in $E\left(\mathbb{F}_q\right)$ then

$$E\left[d\right] \simeq \mathbb{Z}/d\mathbb{Z} \times \mathbb{Z}/d\mathbb{Z}$$

so by basic group theory we are done.

### 3.2 (b)

# 4 A short proof that $\mathbb{Q}[x]/(f_n)$ has no totally real subfields

**Lemma 4.1.** *If $f \in \mathbb{Q}[x]$ has a unique real root that then $\mathbb{Q}[x]/(f)$ has no nontrivial totally real subfield.*

*Proof.* Consider $\overline{\mathbb{Q}}$ an algebraic closure of $\mathbb{Q}$ as a subfield of $\mathbb{C}$, let $G_{\mathbb{Q}}$ be the total Galois group $\mathrm{Gal}\left(\overline{\mathbb{Q}}/\mathbb{Q}\right)$ and let $\sigma \in G_{\mathbb{Q}}$ be the element induced by the complex conjugation on $\mathbb{C}$. Assume towards contradiction that $\iota : K \hookrightarrow \mathbb{Q}[x]/(f)$ is an embedding of a totally real field $K \neq \mathbb{Q}$. Then the restriction induced restriction map

$$\iota^* : \mathrm{Hom}_{\mathbb{Q}}\left(\mathbb{Q}[x]/(f), \overline{\mathbb{Q}}\right) \to \mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$$

is a surjective $G$-equivariant map. The totally realsness of $K$ implies that $\sigma$ acts trivially on $\mathrm{Hom}_{\mathbb{Q}}\left(K, \overline{\mathbb{Q}}\right)$, and hence, the $G$-equivariance of $\iota^*$ implies that $\sigma$ reduces to an action on each fiber of $\iota^*$. However, the fact that $f$ has a unique real root implies that it has an odd degree, and hence, each fiber of $\iota^*$ has an odd cardinality. In particular, since

$\sigma$ is an involution, it must have at least one fixed point in each fiber, which contradicts the fact that $f$ has a unique real root. $\square$

**Corollary 4.2.** *For each $n$, the quotient $\mathbb{Q}[x] / \left(x^n + x^{n-1} + \cdots + x - 1\right)$ has no totally real subfields other than $\mathbb{Q}$.*

## 4.1 Generalization to the finite case

I will try to generalize it to the same generalization as in [1, Appendix A]

**Lemma 4.3.** *Let $E \leq, F, K \leq \overline{E}$ be field extensions such that $E \leq F$ is finite. Assume that there exists a prime $p$ such that $p \nmid [F : E]$ and such that $Aut\left(\overline{E}, K\right)$ is a pro-$p$ group. Then for any totally $K$ sub-extension $E \leq L \leq F$,*

$$[L : E]_s \leq \left|\mathrm{Hom}_E\left(F, K^{ins}\right)\right|$$

*Proof.* Let $G$ be the Galois group $\mathrm{Gal}\left(\overline{E}/E\right)$. Let $\iota : L \hookrightarrow F$ be an $E$-embedding of a totally $K$ extension. Consider the restriction map

$$\iota^* : \mathrm{Hom}_E\left(F, \overline{E}\right) \to \mathrm{Hom}_E\left(L, \overline{E}\right)$$

Since $L$ is totally $K$, $\mathrm{Aut}\left(\overline{E}/K\right)$ acts trivially on $\mathrm{Hom}_E\left(L, \overline{E}\right)$. Hence, the $G$-equivariance of $\iota^*$ implies that $G$ acts (continuously) on each fiber of $\iota^*$, and in particular, so does the pro-$p$ group $\mathrm{Aut}\left(\overline{E}/K\right)$. However, the cardinality of each fiber of $\iota^*$ divides $[F : E]$ and hence is prime to $p$. Therefore, each fiber of $\iota^*$ has a fixed point under the action of $\mathrm{Aut}\left(\overline{E}/K\right)$, i.e. an element of $\mathrm{Hom}_E\left(F, K^{\mathrm{ins}}\right)$. Therefore,

$$[L : E]_s = \left|\mathrm{Hom}_E\left(L, \overline{E}\right)\right| \leq \left|\mathrm{Hom}_E\left(F, K^{\mathrm{ins}}\right)\right|$$

as required. $\square$

**Remark 4.4.** *This is in fact a simplification of the main part of the proof of [1, Theorem A.5]. So I guess if we copy it and paste instead of that part, the entire appendix can be slightly shortened.*

## 4.2 Side Remarks

**Remark 4.5.** *In [1, p. 60] it is claimed that given a directed set $I$ and a function $f : I \to [0, \infty)$, there is at least one limit point in $[0, \infty]$. Indeed it follows from the following slightly more general standard fact:*

**Lemma 4.6.** *Let $X$ be a compact topological space and let $f : I \to X$*

*be a function from a directed poset $I$ to $X$. Then there exists at least one limit point of $f$ in $X$.*

*Proof.* (Inspired by [4, Ch. 3 Supplementary Exercisese, question 10]) For each $i \in I$, let

$$B_i = \overline{\{f(j) \mid j \geq i\}}$$

be the closure of the image of the upper cone of $i$. $B_i$ clearly has the finite intersection property and hence by compactness of $X$, $\bigcap_{i \in I} B_i \neq \emptyset$. Let $x$ be an element in this intersection and let $U$ be an arbitrary open neighborhood of $x$. Then for each $i \in I$, $x \in B_i$ and every open neighborhood of $x$ intersects $B_i$. In particular, $U$ intersects $B_i$ and hence there exists some $j \geq i$ such that $f(j) \in U$. Thus, we have shown that for any open neighborhood $x \in U$, $f^{-1}(U)$ is cofinal in $I$ and hence, $x$ is a limit point of $f$. □

**Remark 4.7.** *According to [1, Proposition A.4], gien a directed poset $I$ and inverse system of finite sets $\{A_i\}_{i \in I}$,*

$$\# \varprojlim_{i \in I} A_i \leq \liminf_{i \in I} \# A_i$$

*Is there an example where the inequality is strict? The answer is yes.*

*Assume* $I = \mathbb{N}$ *and*

$$A_n = \begin{cases} \{1\} & n \text{ odd} \\ \{1,2\} & n \text{ even} \end{cases}$$

*and for any* $m < n$,

$$\varphi_{n,m}(A_n) = \{1\}$$

*Moreover, we can even build a similar example in the context of [1, Appendix A].*

# 5 Background

**Lemma 5.1.** *Let* $f \in \mathbb{Z}[x]$, *and let* $L$ *be its splitting field over* $\mathbb{Q}$. *Assume that over* $\mathbb{F}_p$, *the polynomial* $\overline{f}$ *decomposes as*

$$\overline{f}(x) = \prod_{i=1}^{k} \overline{f_i}(x)^{e_i}$$

*where the* $\overline{f_i}$ *are distinct monic irreducible polynomials. Let* $\mathfrak{P}$ *be a prime of* $\mathcal{O}_L$ *lying over* $(p) \subseteq \mathbb{Z}$. *Then there is a partition, of the set of roots of* $f$ *into disjoint subsets* $S_1, \ldots, S_k$ *such that:*

1. *for each* $1 \leq i \leq k$, $|S_i| = e_i \deg\left(\overline{f_i}\right)$

2. *The map*

$$\pi_{\mathfrak{P}} : \mathcal{O}_L \to \mathcal{O}_L/\mathfrak{P}$$

*reduces to a surjective* $e_i$ *to* 1 *map from* $S_i$ *onto the set of roots of* $\overline{f_i}$ *over* $\mathbb{F}_p$.

3. *The decomposition group at* $\mathfrak{P}$, $D(\mathfrak{P})$, *preserves each* $S_i$

4. *The decomposition group at* $\mathfrak{P}$ *acts on* $S_i$ *in a way that respects the partition into fibers of* $\pi_{\mathfrak{P}}$.

5. *The inertia group at* $\mathfrak{P}$ *acts trivially on the set of fibers of* $\pi_{\mathfrak{P}}$.

6. *The preimage of the frobenious automorphism* $\phi : x \mapsto x^p$ *under* $\pi_{\mathfrak{P}}$ *acts on* $S_i$ *that respects the partition into fibers of* $\pi_{\mathfrak{P}}$, *and induces on the set of fibers a cycle of length* $\deg\left(\overline{f_i}\right)$.

*Proof.* By Hensel's lemma there are $f_1, \ldots, f_k \in \mathbb{Z}_p[x]$ such that for each $1 \leq i \leq k$, $\left(\overline{f_i}\right)^{e_i}$ is the image of $f_i$ under the cannonical map $\mathbb{Z}_p[x] \to \mathbb{F}_p[x]$, and

$$f(x) = \prod_{i=1}^{k} f_i(x)$$

over $\mathbb{Z}_p$. Set $S_i$ to be the set of roots of $f_i$ in $L$.

Fix $1 \leq i \leq k$ and let $\beta_1, \ldots, \beta_{\deg\left(\overline{f_i}\right)}$ be the roots of $\overline{f_i}$ over $\mathbb{F}_p$. Then

$$\begin{aligned}
&\prod_{\alpha \in S_i} \left(x - \pi_{\mathfrak{P}}(\alpha)\right) = \\
&\pi_{\mathfrak{P}}\left(\prod_{\alpha \in S_i} (x - \alpha)\right) = \\
&\pi_{\mathfrak{P}}\left(f_i(x)\right) = \\
&\overline{f_i}(x)^{e_i} = \\
&\prod_{j=1}^{\deg\left(\overline{f_i}\right)} (x - \beta_j)^{e_i}
\end{aligned} \tag{2}$$

and hence, since $\mathbb{F}_p^{\mathrm{alg}}[x]$ is a unique factorization domain, for each $1 \leq j \leq \deg\left(\overline{f_i}\right)$, there are exactly $e_i$ elements $\alpha \in S_i$ such that $\pi_{\mathfrak{P}}(\alpha) = \beta_j$. This proves (2) and in particular it implies (1).

Let $\sigma \in D(\mathfrak{P})$. Then $\sigma$ extends to an automorphism $\widehat{\sigma}$ of the completion of $L$ with respect the $\mathfrak{P}$-valuation $v_{\mathfrak{P}}$. We shall denote this completion by $\widehat{L}$. Since $\widehat{\sigma}$ fixes $\mathbb{Q}$, it also fixes its closure with respec to $v_{\mathfrak{P}}$, a closure which is also known as $\mathbb{Q}_p$. In particular, the automorphism it induces on $\widehat{L}[x]$ fixes $\mathbb{Z}_p[x]$ and thus it fixes $f_i$ and consequently so does the automorphism that $\sigma$ it induces $L[x]$.

Therefore,

$$\prod_{\alpha \in S_i} (x - \sigma(\alpha)) =$$

$$\sigma\left(\prod_{\alpha \in S_i} (x - \alpha)\right) =$$

$$\sigma(f_i) =$$

$$f_i =$$

$$\prod_{\alpha \in S_i} (x - \alpha)$$

and hence, since $L[x]$ is a unique factorization domain, it follows that $\sigma(S_i) = S_i$. Since the same is true for any $i$ and for any $\sigma \in D(\mathfrak{P})$, it follows that $D(\mathfrak{P})$ preserves each $S_i$ and.

It is well known that each element of the decomposition group induces an automorphism of the $L/\mathfrak{P}$, which in particular implies that it respects the partition of $L$, and in particular of $S_i$ into fibres of $\pi_{\mathfrak{P}}$. The automorphism that any element of the inertia group, $I(\mathfrak{P})$, induces on $L/\mathfrak{P}$ is the identity, by the mere definition of the inertia group, and hence the bijection that any such automorphism induces from the set of fibres to iteself is also the identity.

By a slight abuse of notation, use $\phi$ to denote some member of $D(\mathfrak{P})$ that induces the frobenious map $\phi$ on $L/\mathfrak{P}$ (it is a well known fact that any automorphism of $L/\mathfrak{P}$ is induces by some automorphism of $L$). Then as we have already shown, sicne it is a member of $D(\mathfrak{P})$ its action on $S_i$ preserves the partition into fibres of $\pi_{\mathfrak{P}}$. Moreover, the action it induces on them is the action it induces on their images in $L/\mathfrak{P}$ which is indeed a cycle of order

$$[L/\mathfrak{P} : \mathbb{F}_p] = \deg\left(\overline{f_i}\right)$$

$\square$

**Definition 5.2.** *In the situation of lemma 5.1, we will call the each $S_i$ a decomposition block of $\mathfrak{P}$ and we will call the intersection of each $S_i$ with each fibre under $\pi_{\mathfrak{P}}$ an inertia block of $\mathfrak{P}$.*

# 6 Reduction to a 2-adic Question

For each $n \in \mathbb{Z}_{>1}$ denote

$$f_n(x) = x^{n-1} - \sum_{i=0}^{n-2} x^i$$

$$g_n(x) = (x-1) f_n(x) = x^n - 2x^{n-1} + 1$$

We will ommit for now the proof [or even any reference to that fact] that $f_n$ is irreducible over $\mathbb{Q}$. Instead, we will reduce the question of whether its Galois group is the full symmetric group to a 2-adic question. Denote by $L_n$ the splitting field of $f_n$ over $\mathbb{Q}$ and by $G_n = \mathrm{Gal}(L_n/\mathbb{Q})$ its Galois group.

**Lemma 6.1.** *Let $\mathfrak{P} \subseteq \mathcal{O}_{L_n}$ be a prime at which $f_n$ ramifies that lies over an odd prime $(p) \subseteq \mathbb{Z}$. Then the inertia group of $f_n$ at $\mathfrak{P}$ is generated by a single transposition.*

*Proof.* It suffices to show that over $\mathbb{F}_p$, the polynomial $f_n$ has exactly one double root and all other roots are simple. Indeed, assume that this is the case and denote by $\alpha$ the unique double root of $f_n$ over $\mathbb{F}_p$. Then, any other root $\beta$ of $f_n$ over $\mathbb{F}_p$ lifts to a single root, and hence any

element of the inertia group must fix it. Therefore, the inertia group is embedded into the permutation group of the two lifts of $\alpha$, and hence is either trivial or generated by a single transposition. However, since by assumption $f_n$ ramifies at $p$, the inertia group is non-trivial, and hence it is indeed generated by the transposition of the two lifts of $\alpha$.

We shall show then that $f_n$ has exactly one double root over $\mathbb{F}_p$ and all other roots are simple. If $\alpha$ is a multiple root of $f_n$ over $\mathbb{F}_p$, it is in particular a multiple root of $g_n$ and hence

$$0 = g_n'(\alpha) = n\alpha^{n-1} - 2(n-1)\alpha^{n-2}$$

and hence, since $\alpha \neq 0$,

$$\alpha = \frac{2(n-1)}{n}$$

and it is in particular unique.

Since $f_n$ is by assumption ramified at $p$, such $\alpha$ exists. We will show that the multiplicity of $\alpha$ is not greater than 2. It suffices to show that $g_n$ has no root of multiplicity greater than 2, and since all of its roots are non-zero, its multiset of roots is obtained from the multiset of $h_n(x) = x^n - 2x + 1$ by inversion, and hence it suffices

to show that $h_n$ has no root of multiplicity greater than 2 over $\mathbb{F}_p$. However, any root of multiplicity greater than 2 must also be a root of $h_n''(x) = n(n-1)x^{n-2}$, and hence must be 0, which is not a root of $h_n$. $\square$

**Lemma 6.2.** *Let $L/\mathbb{Q}$ be a Galois extension. Then*

$$G = \sum_{\mathfrak{P} \subseteq \mathcal{O}_L} I_{\mathfrak{P}}$$

*where $I_{\mathfrak{P}}$ is the inertia group at the prime $\mathfrak{P}$.*

*Proof.* The only unramified extension of $\mathbb{Q}$ is $\mathbb{Q}$ itself, and hence the fixed field of $\sum_{\mathfrak{P}} I_{\mathfrak{P}}$ is $\mathbb{Q}$. $\square$

**Lemma 6.3.** *A transitive permutation group generated by transpositions is the full symmetric group.*

*Proof.* Sketch inspired by Itamar Israeli: There is a seuqnece of transpositions that connects any two elements. Given a product of a pair of transpositions

$$(a\ b)(b\ c) = (a\ b\ c)$$

it can be shoretned to

$$(b\ c)\,(a\ b)\,(b\ c) = (a\ c)$$

$\square$

**Corollary 6.4.** *If $n$ is odd, then $G_n = S_{n-1}$.*

**Corollary 6.5.** *Assume that $n$ is even. To prove that $G_n = S_{n-1}$, it suffices to show that for any prime $\mathfrak{P} \subseteq \mathcal{O}_{L_n}$ above 2, the inertia group $I_{\mathfrak{P}}$ is generated by a transposition.*

We can also derive a slightly different reduction.

**Lemma 6.6.** *A primitive transitive permutation group $G$ containing a transposition is the full symmetric group.*

*Proof.* Since it is transitive and primitive, any non-trivial normal subgroup of $G$ is also transitive. Indeed the orbits of any normal subgroup form a partition of the set preserved by $G$. Therefore, the normal subgroup generated by the transposition whose existence is assumed is transitive, and hence must be the full symmetric group, and it is generated by all the conjugates of that transposition, which are all transpositions in themselves. $\square$

**Lemma 6.7.** *If $n > 2$ then $d_n$ is not a power of 2.*

*Proof.* By [3],

$$d_n = \pm \frac{n^n - 2^n (n-1)^{n-1}}{(n-2)^2}$$

If $n$ is odd, then $n^n - 2^n (n-1)^{n-1}$ is odd as well, and hence $d_n$ is not a power of 2. Assume that $n$ is even, then by Lemma 7.3,

$$v_2(d_n) = n - 2$$

□

**Corollary 6.8.** *To prove that $G_n = S_{n-1}$, it suffices to show that $G_n$ is primitive and that the discriminant of $L_d$ over $\mathbb{Q}$ is not a power of 2 (for which is sufficient to show that $d_n$ is not a power of 2).*

**Lemma 6.9.** *A permutation group $G$ that contains an $n$-cycle and a transposition is symmetric.*

*Proof.* I guess we could use a version of Itamar's proof here as well □

# 7 Random Facts

## 7.1 The Polynomial is irreducible

**Lemma 7.1.** *$f_n$ is irreducible over $\mathbb{Q}$.*

*Sketch:* Assume $n > 2$ (otherwise it is trivial). The winding number of $g_n$ as $x$ traverses the circle of radius $1+\epsilon$ is $n-1$ for sufficiently small $\epsilon > 0$. Indeed, for such $\epsilon$,

$$|2x^{n-1}| > |x^n| + 1 \geq |x^n + 1|$$

since

$$(1+\epsilon)^n - 2(1+\epsilon)^{n-1} + 1 < 0$$

since

$$g'(1) = n - 2(n-1) = 2 - n < 0$$

Therefore, it has $n-1$ roots of absolute value at most 1. One of these roots is 1, and hence $f_n$ has $n-2$ roots of absolute value at most 1.

Moreover, the only root of absolute value exactly 1 is 1 itself. Indeed, it suffices to show that the only root of $\hat{g_n}(x) := g_n(1/x)\, x^n =$

$x^n - 2x + 1$ of absolute value 1 is 1 itself (since the roots of $\hat{g_n}$ are the inverses of the roots of $g_n$). However, any algebraic number $\alpha$ of absolute value 1 is a root of unity, and hence if $g_n(\alpha) = 0$ then $1 - 2\alpha$ is a root of unity as well, and in particular, by the inverse triangle inequality,

$$1 = |2\alpha - 1| \geq 2 - 1 = 1$$

and hence $\alpha = 1$.

Therefore, $f_n$ has $n - 2$ roots of absolute value strictly less than 1. In other words, only one of its roots has absolute value greater than or equal to 1 (in fact greater). If we assume towards contradiction that $f_n$ is reducible over $\mathbb{Q}$, then since it is monic, it is reducible over $\mathbb{Z}$ as well, and hence one of its factors is a monic polynomial with integer coefficients whose roots are all of absolute value strictly less than 1 which is a contradiction (because then for example the norm, which is the free coefficient up to a sign, is less than 1 in absolute value). $\square$

## 7.2 The Action of the Inertia Group on the Roots of $f_n$

**Lemma 7.2.** *Let $\mathfrak{P}$ be a prime of $\mathcal{O}_{L_n}$ lying above $(2) \subseteq \mathbb{Z}$. Then the inertia group $I_{\mathfrak{P}}$ acts transitively on each inertia block of $\mathfrak{P}$.*

*Proof.* Tweak the proof to apply to the tamely ramified case as well

Let $B$ be an inertia block of $\mathfrak{P}$. It suffices to show that all elements of $B$ are conjugate over $\mathbb{Q}_2^{\mathrm{un}}$, the maximal unramified extension of $\mathbb{Q}_2$. To show that in turn, it suffices to show that the degree of one of these elements over $\mathbb{Q}_2^{\mathrm{un}}$ is at least the size of $B$. Indeed, assume that $\alpha \in B$ and assume that the degree of $\alpha$ over $\mathbb{Q}_2^{\mathrm{un}}$ is at least $e = |B|$. Assume that $\beta$ is a conjugate of $\alpha$ over $\mathbb{Q}_2^{\mathrm{un}}$. Then there is an automorphism $\sigma$ of $\mathbb{Q}_2^{\mathrm{alg}}$ over $\mathbb{Q}_2^{\mathrm{un}}$ such that $\sigma(\alpha) = \beta$. In particular, it reduces to a member of the inertia subgroup of $\mathrm{Gal}(L_n/\mathbb{Q})$ at $\mathfrak{P}$ that maps $\alpha$ to $\beta$ and hence $\alpha$ and $\beta$ are equivalent modulo $\mathfrak{P}$ and hence $\beta \in B$. Therefore, for the number of conjugates of $\alpha$ over $\mathbb{Q}_2^{\mathrm{un}}$ to reach $e$, all elements of $B$ must be conjugate to $\alpha$ over $\mathbb{Q}_2^{\mathrm{un}}$.

It remains to show that there is a root $\alpha \in B$ whose degree over $\mathbb{Q}_2^{\mathrm{un}}$ is at least $e$ or equivalently that there is $\beta \in \mathbb{Q}_2^{\mathrm{un}}(\alpha)$ whose 2-adic

valuation is in the interval $(0, 1/e]$. Indeed, let $k = n/2^e$ and let $\beta$ be the unique primitive $k$-th root of unity whose image under $\pi_{\mathfrak{P}}$ is the image of $B$ under $\pi_{\mathfrak{P}}$. Then the free coefficient of $g_n(x+\beta)$ is

$$g_n(\beta) = \beta^n - 2\beta^{n-1} + 1 = 2\left(1 - \beta^{n-1}\right)$$

whose 2-adic valuation is exactly 1. Therefore, since $g_n(x+\beta)$ has exactly $2^e$ roots whose $\mathfrak{P}$-valuation is strictly positive, by Newton's polygon argument, there is a root $\alpha$ of $g_n(x+\beta)$ whose $\mathfrak{P}$-valuation is not more than $1/2^e$. $\square$

## 7.3 The Discriminant of $f_n$

**Lemma 7.3.** *Assume that $n \geq 3$ is even then*

$$v_2(d_n) = n - 2$$

*Proof.* By [3]

$$d_n = \frac{(-1)^{\binom{n}{2}} \left(n^n - 2^n (n-1)^{n-1}\right)}{(n-2)^2} = (-1)^{\binom{n}{2}} 2^n \cdot \frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{(n-2)^2}$$

and hence if we denote $m = v_2(n-2)$ we get that

$$v_2(d_n) = n - 2m + v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right)$$

Therefore, it suffices to show that

$$v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right) = 2m - n + n - 2 = 2m - 2$$

Assume that $m \geq 2$ Indeed,

$$n - 2 \equiv \epsilon 2^m \mod 2^{2m-1}$$

for some odd $\epsilon$, and hence

$$n \equiv \epsilon 2^m + 2 \mod 2^{2m-1}$$

so

$$
\begin{aligned}
&\left(\frac{n}{2}\right)^n - (n-1)^{n-1} \equiv \\
&\left(\epsilon 2^{m-1}+1\right)^n - \left(\epsilon 2^m+1\right)^{n-1} \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + n\epsilon 2^{m-1} + 1 - (n-1)\,\epsilon 2^m - 1 \equiv \backslash 3m-3 \geq 2m-1 \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(\frac{n}{2} - (n-1)\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(\epsilon 2^{m-1} + 1 - \epsilon 2^m - 1\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} + \epsilon 2^m\left(-\epsilon 2^{m-1}\right) \equiv \\
&\binom{n}{2}\epsilon^2 2^{2m-2} - \epsilon^2 2^{2m-1} \equiv \\
&\left(\binom{n}{2} - 2\right)\epsilon^2 2^{2m-2} \mod 2^{2m-1}
\end{aligned}
$$

and in particular its $v_2$ is $2m-2$ as required.

Assume finally now that $m = 1$, then

$$v_2\left(\frac{n}{2}\right) \geq 1$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n\right) \geq n > 0 = v_2\left((n-1)^{n-1}\right)$$

and hence

$$v_2\left(\left(\frac{n}{2}\right)^n - (n-1)^{n-1}\right) = 0 = 2m - 2$$

Assume that $m = 2$, then

$$v_2\left(n^n\right) \geq 2n > v_2\left(2^n (n-1)^{n-1}\right)$$

and hence

$$v_2\left(n^n - 2^n (n-1)^{n-1}\right) = n$$

and hence

$$v_2\left(d_n\right) = n - 2$$

□

**Corollary 7.4.** *There is an odd prime that divides $d_n$ for any $n \geq 4$.*

*Proof.* Sketch: The maximal odd divisor of $d_n$ is

$$\frac{n^n - 2^n (n-1)^{n-1}}{2^{v_2\left(n^n - 2^n (n-1)^{n-1}\right)} (n-2)^2} =$$
$$\frac{n^n - 2^n (n-1)^{n-1}}{2^{n-2} (n-2)^2} =$$
$$\frac{\left(\frac{n}{2}\right)^n - (n-1)^{n-1}}{\left(\frac{n-2}{2}\right)^2}$$

and the numerator clearly clearly grows faster than the denominator.

□

### 7.4 Ramification at 2

**Lemma 7.5.** *Let $n \geq 4$ be even. Then $L_n$ ramifies at 2.*

*Proof.* Let $v$ be a valuation extending $v_2$ to $L_n$ and let $\widehat{L}$ be the completion of $L_n$ with respect to $v$. It suffices to show that □

## 8 Random Calculations

### 8.1 The case $(n = 3)$

This case is the just the classic Golden Ratio calculation. Explicitly,

$$f_3(x) = x^2 - x - 1 = 0$$

its roots are given by

$$\alpha = \frac{1+\sqrt{5}}{2}, \quad \beta = \frac{1-\sqrt{5}}{2}$$

The difference between them is $\sqrt{5}$ which indeed generates the unique prime ideal above 5 in $\mathcal{O}_{\mathbb{Q}(\sqrt{5})}$ and thus the unique element of $G(f_3)$ is ramified above 5 as expected. Explicitly, both solutions are mapped to $3 \in \mathbb{F}_5$ and hence the unique of $G(f_3)$ acts trivially on the residue field.

In a more modern (but maybe in this case redundant) language, we could note that $\alpha$ generates $\mathcal{O}_{\mathbb{Q}_5(\sqrt{5})}$ over $\mathcal{O}_{\mathbb{Q}_5}$. Indeed, following the procedure presented in Serre's proof of [9, III §6 Proposition 12], we can note that

$$v_5(\alpha + 2) = v_5\left(\frac{5+\sqrt{5}}{2}\right) = 1$$

and hence

$$1, \alpha, \alpha + 2, \alpha^2 + 2\alpha$$

generate $\mathcal{O}_{\mathbb{Q}_5(\sqrt{5})}$ over $\mathcal{O}_{\mathbb{Q}_5}$ as a module [maybe I need to show it explicitly at some point]. Hence the fact that the unique element of $G(f_3)$ maps $\alpha$ to $\beta$ which is in the same $\left(\sqrt{5}\right)$ coset, implies that it belongs to the ramification group.

## 8.2 The case $(n = 5)$

First off, a direct calculation shows that

$$d_5 = \pm 563$$

in particular, it ramifies only above 563. Moreover, in $\mathbb{F}_{563}$, the unique double root is

$$\alpha := \frac{2 \cdot 4}{5} \equiv 8 * 338 \equiv 452 \mod 563$$

## 8.3 The Case $n = 10$

Let $L$ be the splitting field of $f_{10}$ over $\mathbb{Q}_2$.

**Lemma 8.1.** *$L$ is ramified over* $(2)$.

*Proof.* It suffices to show that $\mathcal{O}_L$ contains an element of fractional valuation. Note that over $\mathbb{F}_2$,

$$\overline{g_{10}}(x) = x^{10} + 1 = \left(x^5 + 1\right)^2 = (x+1)^2 \left(x^4 + x^3 + x^2 + x + 1\right)^2$$

Therefore, the roots of $g_{10}$ over $\mathbb{Q}_2$ are given by two lifts of 1 and two

lifts of each root of $x^4+x^3+x^2+x+1$ over $\mathbb{F}_{2^4}$. Choose $\overline{\beta}$ to be a root of $x^4+x^3+x^2+x+1$ in $\mathbb{F}_{2^4}$. Let $\beta \in \mathcal{O}_L$ be a lift of $\overline{\beta}$ (not necessarily a root of $g_{10}$). Consider the polynomial,

$$f_{10}\left(x+\beta\right)$$

Its roots are given by $\alpha_i - \beta$ where $\alpha_i$ are the roots of $f_{10}$. Therefore, if we fix a prime $\mathfrak{P} \subseteq \mathcal{O}_L$ lying over (2), by Lemma 5.1, $f_{10}\left(x+\beta\right)$ has exactly two roots in $\mathfrak{P}$. $\square$

We will show that $G_{10}$ is primitive. Let $\alpha_1, \ldots, \alpha_9$ be the roots of $f_{10}$. Then in $\mathbb{F}_2$,

$$\begin{aligned}
&\overline{(x-1)\prod_{i=1}^{9}\left(x-\alpha_i\right)} = \\
&\overline{g_{10}}\left(x\right) = \\
&x^{10}-2x^9+1 = \\
&x^{10}+1 = \\
&\left(x^5+1\right)^2 = \\
&\left(x+1\right)^2\left(x^4+x^3+x^2+x+1\right)^2
\end{aligned}$$

## 8.4 The case $n = 2p$ for $p$ an odd prime

We will show that $G_n$ is primitive. Assume towards contradiction that there is a partition of the $n-1$ roots of $f_n$ into blocks respceted by $G_n$. Since $G_n$ acts transitively on the roots, all blocks must be of the same size, say $d$, and denote the blocks by $B_1, \ldots, B_k$ where $k = \frac{n-1}{d}$.

Note now that over $\mathbb{F}_2$, $g_n$ factors as

$$g_n = x^n + 1 = (x^p + 1)^2 = (x+1)^2 \left( \sum_{i=0}^{p-1} x^i \right)^2$$

and hence

$$\overline{f_n} = \overline{\frac{g_n}{x-1}} = (x+1) \left( \sum_{i=0}^{p-1} x^i \right)^2$$

Fix a prime $Q$ of $\mathcal{O}_{L_n}$ lying over (2). Let $\alpha$ be the unique root of $f_n$ that projects to 1 and assume WLOG that $\alpha \in B_1$. Since it forms its own decomposition block, the decomposition group at $Q$ stabilizes $\alpha$. In particular, it stabilizes $B_1$. Therefore, since by lemma [will be added later], the inertia group at $Q$ acts transitively on each inertia block, the rest of $B_1$ is a union of $\frac{d-1}{2}$ inertia blocks. Therefore, since the frobenious automorphism at $Q$ acts on the set of inertia blocks as

a cycle of length $\mathrm{ord}_p(2)$, it follows that $\mathrm{ord}_p(2) \mid \frac{d-1}{2}$.

Now, fix $1 < i \leq k$. Since $\#B_i = d$ is odd, $B_i$ cannot be a union of inertia blocks, and hence it must contain at least one root, $\alpha$ such that there is at least one automorphism $\sigma$ that maps $\alpha$ to a root in $B_j$ for some $j \neq i$. However, since $G$ respects the block structure, $\sigma$ must map the entire block $B_i$ to $B_j$. Therefore, for each root $\beta \in B_i$, the unique other root that shares its inertia block must also be in $B_j$.

Let $\phi \in D(\mathfrak{P})$ be the frobenious. Then one of two options must hold:

1. $\phi(B_i) = B_i$. But this implies that $\mathrm{ord}_p(2) \mid d$ which is impossible since then $\mathrm{ord}_p(2) \mid \gcd\left(d, \frac{d-1}{2}\right) = 1$.

2. There is $\alpha \in B_i$ such that $\phi(\alpha) \in B_j$ for some $j \neq i$.

and denote by $I \leq G_n$ the inertia group corresponding to $Q$. Then the map $\pi_Q$ reduces to a surjective 2 to 1 map from the roots of $g_n$ to the $\mu_{p-1} \subseteq \overline{\mathbb{F}_2}$ and the action of $I$ stabilizes the fibers of this map.

In particular, the set of roots of $g_n$ includes exactly two lifts of 1 which the action of $I$ either fixes or swaps. However, $1 \in \mathbb{Q}$ must be one of these two lifts, and hence $I$ must fix both of them. Denote the

second lift of 1 by $\alpha$ and denote by $B_1$ the block containing $\alpha$. Then since $G_n$ respects the block structure, $I$ must stabilize $B_1$.

By lemma [will be added later], the action of $I$ on each inertia block is transitive. Therefore,

### 8.5 The case $n = 2^d$

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